Course Code: 303-04
Course Title : Network Security and Penetration Testing

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<tr><td>Course Code</td><td>303-04</td></tr>
<tr><td>Course Title</td><td>Network Security and Penetration Testing</td></tr>
<tr><td>Credit</td><td>4</td></tr>
<tr><td>Course Category</td><td>Major</td></tr>
<tr><td>Level of Course</td><td>300-399 (Higher Level)</td></tr>
<tr><td>Teaching per week</td><td>2 Hours Theory + 4 Hours Practical work</td></tr>
<tr><td>Total Hours</td><td>90 Hrs.</td></tr>
<tr><td>Minimum Weeks per Semester</td><td>15 Weeks (including Class work, examination, preparation, Case study etc.)</td></tr>
<tr><td>Implementation Year:</td><td>A.Y. 2025-2026</td></tr>
<tr><td>Medium of Instruction</td><td>English</td></tr>
<tr><td>Purpose of Course</td><td>To gain basic understanding of the concepts of Network Security, its applications and tools.</td></tr>
<tr><td>Course Objective</td><td>(i) Understanding network security principles and threat mitigation.
(ii) Learning firewall and IDS/IPS configurations.
(iii) Conducting penetration testing using industry-standard tools.
(iv) Exploring hardware components involved in network security.</td></tr>
<tr><td>Pre-requisites</td><td>Basic Concepts of Network</td></tr>
<tr><td>Course Outcome</td><td>CO1: To make them understand the concepts of Network security
CO2: To make them understand the Firewalls and ways of network monitoring
CO3: To make them understand wireless security and different encryption standards.
CO4: To make them understand the basics of penetration testing and its various techniques.
CO5: To make them understand Network hardening and Network security configurations and practices.</td></tr>
<tr><td>Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)</td><td>
<table>
<tr><td></td><td>PSO1</td><td>PSO2</td><td>PSO3</td><td>PSO4</td><td>PSO5</td><td>PSO6</td><td>PSO7</td><td>PSO8</td></tr>
<tr><td>CO1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>CO2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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<tr><td>CO4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>CO5</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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<tr><td>Course Content</td><td>Chapter - 1. Fundamentals of Network Security
1.1. Introduction to Network Security Concepts
1.2. OSI & TCP/IP Security Models
1.3. Common Network Threats & Attacks
1.4. Importance of Network Segmentation
1.5. VPNs and Secure Remote Access

Chapter - 2. Firewalls, IDS/IPS, and Network Monitoring
2.1. Firewall Types and Configurations
2.2. Setting Up IDS/IPS (Snort, Suricata)
2.3. Network Traffic Analysis Using Wireshark
2.4. Secure Network Design and Implementation
2.5. Hands-on with pfSense Firewall</td></tr>
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	Chapter - 3. Wireless Security, Hardware, and Encryption Techniques **3.1.** Wi-Fi Encryption Standards (WPA2, WPA3) **3.2.** MAC Filtering and Rogue AP Detection **3.3.** Network Hardware Security (Switches, Routers, Access Points) **3.4.** Securing IoT and Embedded Devices **3.5.** Capturing and Analyzing Wireless Packets **Chapter - 4. Penetration Testing Methodologies & Hardware Exploits** **4.1.** Introduction to Penetration Testing **4.2.** Scanning Networks with Nmap and Nessus **4.3.** Hardware-Based Attacks (Keyloggers, RFID Cloning, USB Exploits) **4.4.** Manual Exploitation vs. Automated Exploitation **4.5.** Generating Penetration Testing Reports **Suggested Practical Exercises:** - Configuring Firewalls and VPNs - Conducting Network Penetration Testing - Securing Wireless Networks - Analyzing Network Logs for Threat Detection - Testing Hardware-Based Attacks and Defense Mechanisms - Writing Network Security Audit Reports
Reference Books	1. Network Security Essentials: Applications and Standards – William Stallings 2. Network Security Bible – Eric Cole 3. Practical Packet Analysis: Using Wireshark to Solve Real-World Network Problems – Chris Sanders 4. Mastering pfSense – David Zientara 5. Applied Network Security Monitoring: Collection, Detection, and Analysis – Chris Sanders & Jason Smith 6. Wireless Security: Models, Threats, and Solutions – Seth Enoka 7. Computer Security – Dieter Gollmann 8. Penetration Testing: A Hands-On Introduction to Hacking – Georgia Weidman 9. Hacking: The Art of Exploitation – Jon Erickson 10. Kali Linux Revealed: Mastering the Penetration Testing Distribution – Raphael Hertzog, Jim O’Gorman, Mati Aharoni
Teaching Methodology	Class Work, Discussion, Self-Study, Case study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.